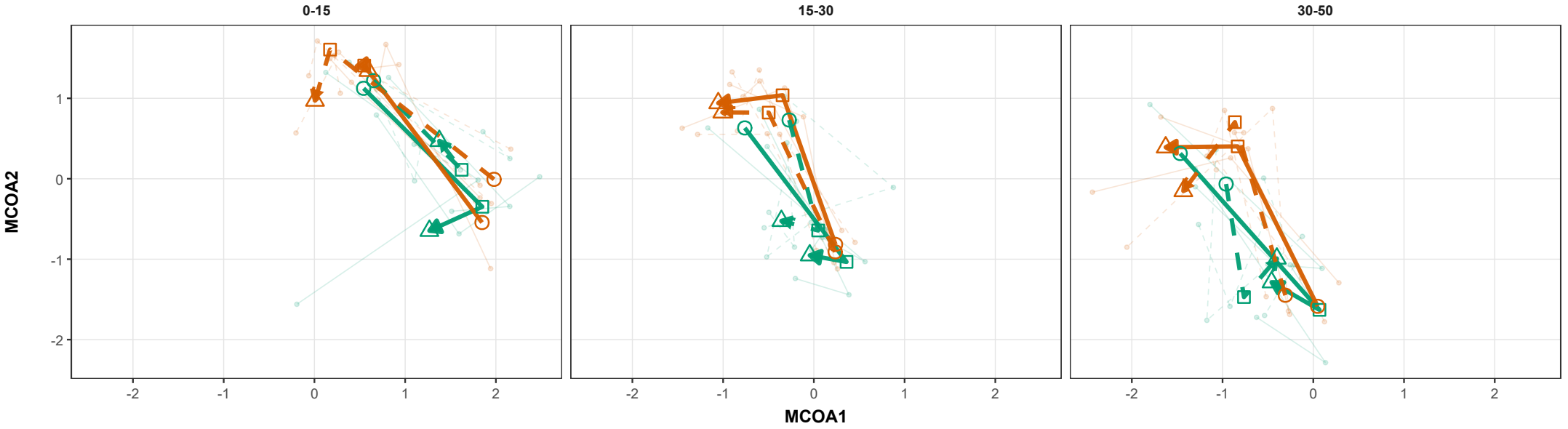
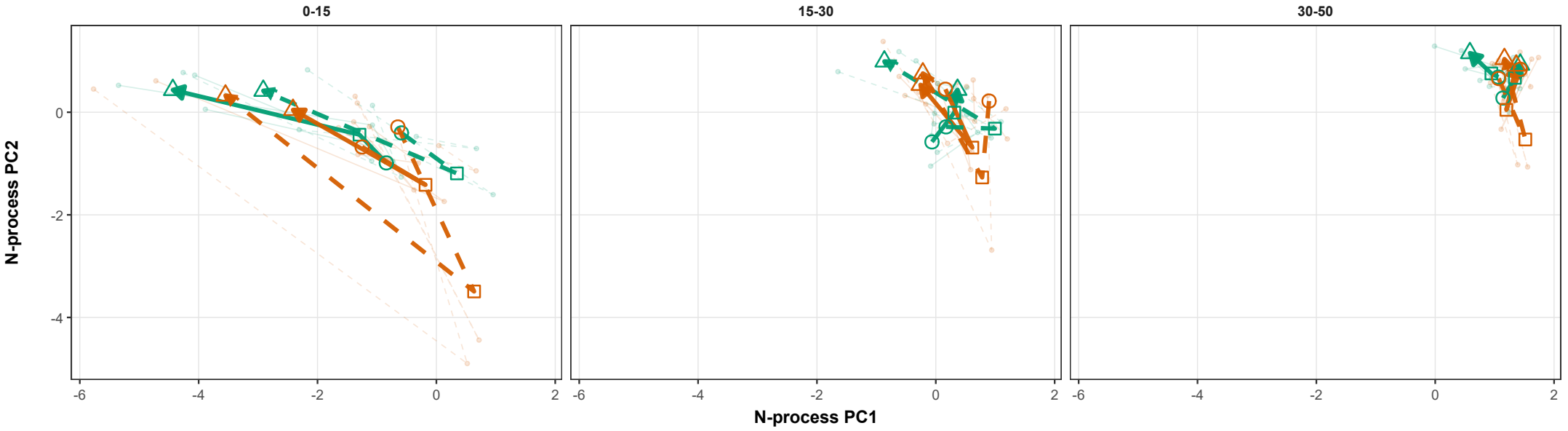
A Multi-kingdom community trajectories

Pale lines: individual units; thick arrows: land-use trajectory × treatment centroids



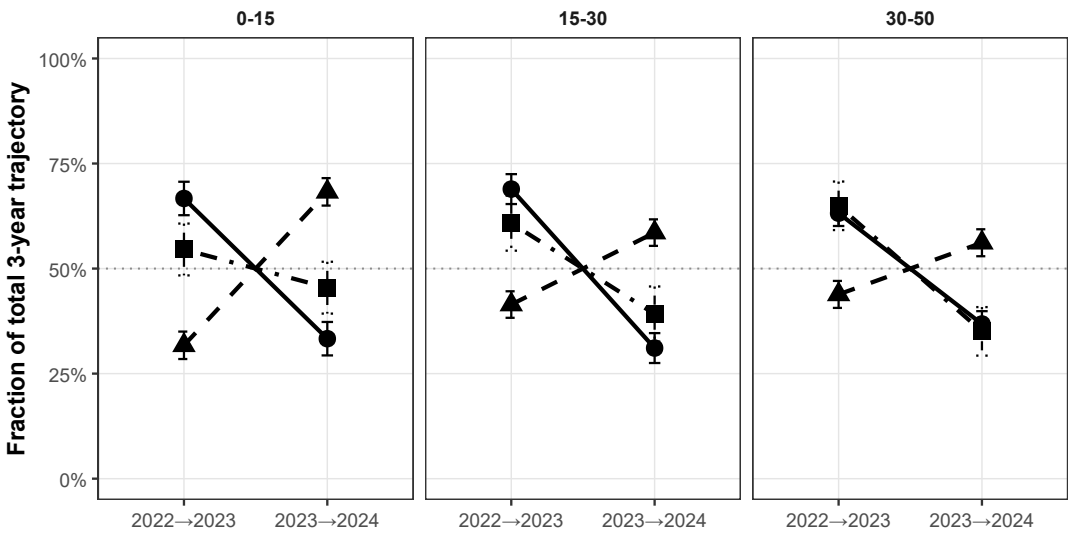
B Potential N-activity trajectories

Pale lines: individual units; thick arrows: land-use trajectory × treatment centroids



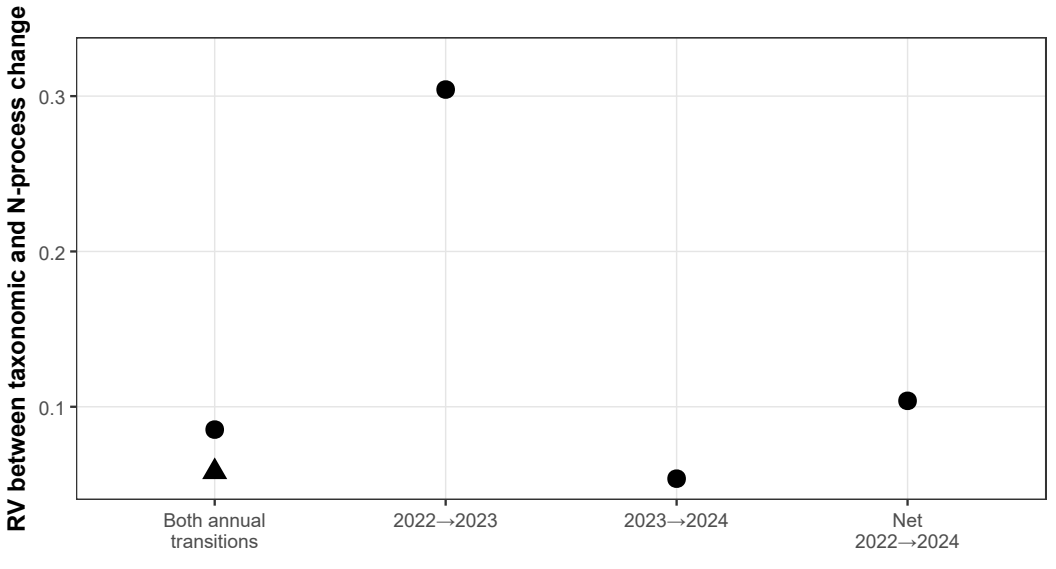
C Temporal ordering across community, gene potential and N activity

Exact whole-profile timing tests: gene vs taxonomy p = 0.274; N activities vs gene p = 0.002



D Temporal covariance

Annual differences are permuted as whole-profile two-interval trajectories; * p < 0.05.



ent Control Heated **Year** 2022 2023 2024 **Land-use trajectory** Stable PP TP -> Crop Taxonomy N gene potential N functioning Design-residual ser